Personal Statement

Yanghao Chen

My interest in large language models led me to my first research project at Tongji University, where I worked with Professor Ying Zhao to fine-tune Qwen for mechanics of materials. Public datasets and Colab notebooks gave me a starting point. To adapt them to the project, I examined how instruction examples were organized, followed the code that loaded them, and prepared my own training data. As I worked through these steps, I began to understand the large Transformer neural network behind the model and how it generated text by predicting successive tokens. Breaking the task into smaller questions helped me connect what I was learning about the model with the code I was using.

Preparing the model for local deployment in the China International College Students' Innovation Competition also involved working closely with teammates. When the model exceeded our available GPU memory, we tested changes to numerical precision and to how it was saved and loaded until it could run locally. Through these discussions, I learned to explain my reasoning more clearly and communicate with my teammates about possible solutions and next steps. The experience of learning from shared resources and working through difficulties with others also influenced how I shared my own work. I published my instruction datasets and fine-tuned models on Hugging Face and documented my projects in public repositories. I wanted other learners to have concrete examples they could study and adapt, as I had when I began.

As I came to understand prediction in language models, I became interested in using neural networks to predict changes in physical systems. This led me to study concentration and stress evolution in battery materials using convolutional long short-term memory (ConvLSTM) models. Following their predictions over successive time steps taught me to examine how errors accumulated during use. Producing the simulation data also led me to explore training without precomputed solution labels, using physics-informed neural networks (PINNs). In a study of the one-dimensional inviscid Burgers equation, I added a fixed-coefficient artificial-viscosity term to the partial differential equation (PDE) residual. The resulting model had a lower relative L^2 error but a higher maximum pointwise error against the analytical inviscid solution. To interpret that difference, I examined the predicted shock and considered how the added viscosity changed the equation being solved. I began paying closer attention to the assumptions behind a result and to what the evidence justified me in claiming.

The need to retrain my PINN when initial or boundary conditions changed drew me toward neural operators, which learn mappings between function spaces. During my visiting research internship at Johns Hopkins University, I explored this approach under Professor Somdatta Goswami's supervision through a hybrid finite element–neural operator project. When a model that performed well on validation data became less accurate during coupling, I compared the interface-displacement inputs and discussed the differences with my collaborator. This helped me identify a mismatch between the training data and the inputs encountered in the coupled solver. Retraining on revised data reduced stress-prediction errors on coupling inputs in cylinder and bracket tests. This internship reinforced my decision to pursue a PhD in computational mechanics and scientific machine learning. I hope to study the integration of numerical methods and learned models for physical simulation in greater depth, strengthen my theoretical foundations, and develop the ability to conduct independent research in this area. I also hope to continue the exchange of ideas, code, and data that has supported my learning.